Caio Lins Peixoto

MSc Student

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Research Interests

Currently working on stochastic control, nonparametric statistics, and infinite dimensional optimization.

Education

School of Applied Mathematics, Getulio Vargas Foundation — FGV EMAp

2024 — Now **MSc in Applied Mathematics and Data Science** GPA: 4.0/4.0.
Thesis: *Gradient-free optimization in infinite dimension with applications to PDEs.*

2020 — 23 **BSc in Applied Mathematics** GPA: 9.8/10.
Thesis: *Nonparametric Instrumental Variable Regression through Stochastic Approximate Gradients.*

Research Experience

2025 — Now **HDI estimation** We are developing a procedure to estimate highest density intervals from (possibly dependent) samples of the target distribution, including asymptotic confidence intervals. Our approach is based on generalized Z-estimation. *Joint work with Luiz Carvalho, Yuri Saporito, Daniel Csillag (FGV EMAp), Flávio Bambirra (UFMG) and Charles Doss (University of Minnesota).*

2024 — Now **Singular stochastic control** We are studying singular stochastic control problems in which the singular control coefficient depends on the current value of the singular control itself. This feature allows the modeling of diminishing transaction costs as the total traded volume increases. *Joint work with Yuri Saporito (FGV EMAp) and Agostino Capponi (Columbia University).*

2024 — Now **Gradient-free optimization in Hilbert spaces** When viewed as the minimization of an L^2 loss over a Sobolev space, PDEs become optimization problems for which exact gradients are extremely costly to compute, but directional derivatives are not. In this work, we build an algorithm which uses directional derivatives to compute infinite-dimensional stochastic gradients over general separable Hilbert spaces, and apply this methodology to PDEs. *Joint work with Yuri Saporito, Bernardo da Costa and Daniel Csillag (FGV EMAp).*

2023 — 24 **NPIV** Co-authored the paper *"Nonparametric instrumental variable regression through stochastic approximate gradients"*. We showed how to formulate a functional stochastic gradient descent algorithm to tackle NPIV regression by directly minimizing the populational risk, obtaining a more flexible and competitive method. Paper link. *Joint work with Yuri Saporito (FGV EMAp) and Yuri Fonseca (UCL).*

2022 **Undergraduate research** Research assistant for Profs. Yuri Saporito and Yuri Fonseca on a project about statistical inverse problems. I implemented the proposed algorithms in Python and proofread the manuscript. The final paper was published in NeurIPS.

Publications

NeurIPS 2024 Fonseca, Yuri, **Peixoto, Caio** & Saporito, Yuri (2024). Nonparametric instrumental variable regression through stochastic approximate gradients. Advances in Neural Information Processing Systems, 37, pp. 131756-131785.

Teaching Experience

2021 — Now **Teaching Assistant:** Linear Algebra, Probability Theory, Real Analysis, Machine Learning (undergrad level); Measure Theory, Linear Algebra, Statistical Inference (grad level). Hosted tutoring sessions, designed and graded homework and exams.

2023 **FGV Amplia** Served as a tutor for FGV's student-led college entrance exam preparation classes for low-income high school students.

Awards and Scholarships

2024 — 2026 **PICME** A funding program organized by the Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES) and IMPA. Awarded **one of 15** national scholarships granted to Brazilian master's students in Mathematics.

2024 **SBMAC** Academic merit prize awarded by the *Brazilian Society for Applied and Computational Mathematics* (SBMAC) to "honor students who stand out for their academic performance in Applied and/or Computational Mathematics in STEM undergraduate courses".

2023 **FGV EMAp** Academic merit prize, for achieving the highest average grade of my cohort.

2020 — 2023 **CDMC** FGV's *Center for the Development of Mathematics and the Sciences*. Full scholarship and a stipend to complete my bachelor's degree in Applied Mathematics at FGV EMAp.

2017 — 19 **OBMEP** *(Brazilian Mathematical Olympiad for Public Schools)*. Two Silver medals and one Bronze medal.

Event Participation

2025 **Presenter** RiO — Research in Options, Rio de Janeiro. Poster "Gradient-freem optimization in infinite dimension with applications to PDEs".

2025 **Presenter** CNMAC — National Congress of Applied and Computational Mathematics, Rio de Janeiro. Poster "Functional gradient descent through directional dderivatives".

2024 **Presenter** NeurIPS, Vancouver, Canada. Poster "Nonparametric instrumental regression through stochastic approximate gradients".

2024 **Presenter** CNMAC, Pernambuco. Poster "Nonparametric instrumental regression through stochastic approximate gradients".

2024 **Organizer** 12th World Congress of the Bachelier Finance Society, FGV EMAp, Rio de Janeiro. Member of the Local Committee.

2023 **Presenter** RiO — Research in Options, Rio de Janeiro. Poster "Nonparametric instrumental regression through stochastic approximate gradients".

2023 **Attender** LACIAM — Latin American Congress on Industrial and Applied Mathematics.

2022 **Attender** CNMAC, Campinas, Brazil.

Skills

Programming Proficient in Python (NumPy, Tensorflow, JAX, PyTorch). Familiar with R, Julia, C++.

Software LaTeX, Git, Linux, Vim.

Languages

Portuguese Native

English TOEFL iBT 113/120